INF273 – Assignment #3

Implement a Local Search, and a Simulated Annealing algorithm to solve a truck and drone delivery problem.

- Work with the same eight instances you used in assignment #2-b
- You should implement the following algorithms, run them 10 times for each instance and report eight tables (one for each problem instance) in the given format
- Start with an initial solution that all customers are served by the truck in the order of their indices, from 1 to N.
- You should also report the best solution found for each instance, using the same format as in your submission for Assignment 2-a. For each instance, report only one solution—namely, the best outcome obtained across runs of the two algorithms.
- You should use the 1-reinsert heuristic as the operator. This means removing one customer from either part one or part two of the solution representation and reinserting it elsewhere. The choice of which customer to remove and where to reinsert it—into either part one or part two—is up to you. Parts three and four should be modified if, after changing parts one and two, they no longer make sense. In fact, it is recommended to modify parts three and four anyway, since without changing them you may not be able to explore different flight possibilities for the drones. Be creative!

Instance name (e.g. R_20)				
	Average objective	Best objective	Improvement (%)	Average running time (in seconds)
Random Search	Copy from A#2	Copy from A#2	Copy from A#2	Copy from A#2
Local Search-1-insert				
Simulated Annealing-1-insert				

* Initial solution: All customers are served by the truck in the order of their indices, from 1 to N.

Local search (modified for assignment #3)

```
Input: initial solution (s_0),
Input: neighborhood operator: 1-reinsert
Input: evaluation function f, f(s) ⟶ the cost of s
BestSolution ⟸ s_0
for iteration = 1 to 10000
    NewSolution ⟸ Operator (BestSolution)
    if NewSolution is feasible and f(NewSolution) < f(BestSolution) then
        BestSolution ⟸ NewSolution
    end if
end for
```

Simulated Annealing (modified for assignment #3)

```
Input: initial solution (s_0),
Input: neighborhood operator: 1-reinsert
Parameters: T_f (final temperature) = 0.1
Input: evaluation function f, f(s) ⟶ the cost of s
Incumbent ⟸ s_0 , BestSolution ⟸ s_0
for w = 1 to 100
    NewSolution ⟸ Operator (Incumbent)
    ΔE ⟸ f(NewSolution) − f(Incumbent)
    if NewSolution is feasible and ΔE < 0 then
        Incumbent ⟸ NewSolution
        if f(Incumbent) < f(BestSolution) then
            BestSolution ⟸ Incumbent
        end if
    elseif NewSolution is feasible
        if Rand < 0.8 then
            Incumbent ⟸ NewSolution
        end if
        Δ_w ⟸ ΔE
    end if
end for
DeltaAvg = mean (Δ_w)
T_0 = −DeltaAvg / ln (0.8) ,  α = ⁹⁹⁰⁰√(T_f / T_0)
T ⟸ T_0
for iteration = 1 to 9900
    NewSolution ⟸ Operator (Incumbent)
    ΔE ⟸ f(NewSolution) − f(Incumbent)
    if NewSolution is feasible and ΔE < 0 then
        Incumbent ⟸ NewSolution
        if f(Incumbent) < f(BestSolution) then
            BestSolution ⟸ Incumbent
        end if
    elseif NewSolution is feasible and Rand < p = e^(−ΔE/T)
        Incumbent ⟸ NewSolution
    end if
    T = α * T
end for
```